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Knowledge-inspired and sample-generation-based spectral augmentation for few-shot underwater acoustic target recognition[a)]

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ABSTRACT:

To address the challenges of data scarcity and non-stationary noise interference in underwater acoustic target recognition in complex marine environments, existing few-shot learning methods often lack effective mechanisms to exploit structured spectrotemporal characteristics under low-signal-to-noise ratio conditions. This paper proposes a sample-generation-based few-shot underwater acoustic target recognition framework with a knowledge-inspired spectral decomposition augmentation strategy and a cross-feature guided recalibration module. The spectral decomposition augmentation module operates in the spatial-frequency domain of spectrograms to construct complementary multi-view samples by preserving coarse spectrotemporal structures while suppressing noise-sensitive fine texture variations. A Siamese network with time-frequency attention is employed to learn robust spectrotemporal representations, while the cross-feature guided recalibration module uses original-view features to perform dynamic gated recalibration on augmented views, thereby reducing sensitivity to noise-induced variations. Multi-view joint learning based on cosine similarity encourages the learning of shared discriminative representations across views. In addition, a task-adaptive fine-tuning mechanism with pseudo-sample generation is introduced to improve adaptation to novel classes. Experimental results on the ShipsEar dataset and the cross-domain DanShip dataset demonstrate that the proposed method achieves superior recognition accuracy and robustness under low-signal-to-noise ratio and cross-domain conditions compared with existing state-of-the-art few-shot learning methods. © *2026 Acoustical Society of America*. https://doi.org/10.1121/10.0044413

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I. INTRODUCTION

Underwater acoustic target recognition (UATR) is a core application of passive sonar technology that plays a critical role in marine environment monitoring and resource exploration. The technology enables automatic classification and identification of marine targets by analyzing ship-radiated noise. As an integral part of sonar signal processing and underwater soundscapes, it is widely used in maritime traffic monitoring, port management, underwater communication, marine environmental acoustic monitoring, and maritime security (Niu *et al.*, 2023). However, physical laws govern underwater acoustic signal propagation, and the complex oceanic hydrographic environment introduces multipath effects and propagation losses. Combined with non-stationary background noise, these factors frequently obscure or suppress the physical characteristics of target signals (Ferguson, 2000). The resulting complexity of the acoustic field significantly degrades signal-to-noise ratios (SNRs), presenting substantial challenges for UATR (Bianco *et al.*, 2019).

Recently, deep learning (DL) has demonstrated powerful capabilities in acoustics and underwater acoustic signal processing. DL methods have not only significantly improved classification accuracy and robustness in UATR tasks but also are increasingly important in sonar signal processing, underwater soundscape analysis, and acoustic environment monitoring. For instance, DL has been successfully applied to typical applications such as sonar image processing (Hong *et al.*, 2025; M. Wang *et al.*, 2025), sonar signal detection (Zhang *et al.*, 2025), underwater target detection and localization (Liu *et al.*, 2023, 2025), and marine biological acoustic identification (Fan *et al.*, 2025; Z. Huang *et al.*, 2025), providing valuable technical references for UATR.

In UATR, researchers have proposed various DL-based recognition models to overcome the limited feature extraction capability of traditional methods. In terms of feature extraction and representation, Yang *et al.* (2024) proposed a UATR method based on feature fusion and residual convolutional neural networks. M. Chen *et al.* (2025) combined

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time-frequency attention with mixture-of-experts mechanisms to construct a lightweight model. Fan *et al.* (2024) extracted multi-level time-frequency features through hierarchical self-attention. Y. Chen *et al.* (2025) also proposed a time-frequency-spatial dual-stream network to enhance spatial feature fusion. For robustness enhancement, Zhou *et al.* (2023) adopted a joint training strategy with a Transformer-based denoising front-end and recognition back-end. Fan *et al.* (2024) proposed dual filtering modules with multi-loss self-distillation training for weak-signal recognition. Y. Wang *et al.* (2025) proposed a frequency-band splitting and weighting network that combines sub-band weighting techniques to improve underwater target recognition accuracy. Fan *et al.* (2025) further used soft-thresholding strategies to reduce non-stationary interference.

However, these methods remain highly dependent on high-quality labeled data. In practical applications, labeled underwater acoustic samples are difficult and costly to acquire at sea (Dong *et al.*, 2021). This problem is particularly severe in deep-sea and offshore environments, where data collection is further constrained by observation equipment, weather conditions, and sea-state variations. Therefore, studying UATR under few-shot conditions is of considerable practical importance.

To address the problem of insufficient training samples, data augmentation techniques have been introduced to increase both the quantity and diversity of training data. Among traditional approaches, D. Li *et al.* (2023) improved data diversity through underwater acoustic channel modeling and transfer learning, while J. Xu *et al.* (2023) combined smoothness-inducing regularization with local mask-copying strategies. Generative-model-based approaches have also been widely explored. Yu *et al.* (2025) utilized the Hilbert–Huang transform together with a deep convolutional generative adversarial network (GAN), while Du *et al.* (2025) combined dictionary learning with multiple discriminative GANs to generate high-quality samples. In addition, Yu *et al.* (2025) integrated diffusion models with audio spectrogram transformers to alleviate data scarcity by generating high-quality Mel-spectrogram samples. Liu *et al.* (2021) further enriched sample generation strategies by combining data augmentation with transfer learning.

However, data augmentation alone often struggles to cover the full range of complex variations in underwater acoustic environments. In contrast, few-shot learning (FSL) enables models to recognize new classes with only a small number of labeled samples, thereby reducing dependence on large-scale annotated datasets. FSL has achieved promising results in image recognition (Liu *et al.*, 2024; Qiao *et al.*, 2018), speech processing (Chen and Yu, 2023; Y. Li *et al.*, 2023), and natural language processing (Ge *et al.*, 2023; Yin, 2020), and has recently been introduced into UATR. In the area of few-shot UATR, W. Huang *et al.* (2025) proposed a multi-task balanced channel-attention convolutional network to improve recognition performance through joint channel attention and multi-task learning. Fu *et al.* (2025) enhanced semantic modeling capability under few-shot conditions through multi-feature fusion and multi-modal modeling. [For cross-domain scenarios, Cui *et al.* (2024) facilitated knowledge transfer from source to target domains through cross-domain contrastive learning and self-supervised training.] Yang *et al.* (2024) developed a method based on Siamese networks and contrastive learning mechanisms. Furthermore, Cui *et al.* (2025) introduced domain adaptation and knowledge distillation methods, Feng *et al.* (2024) employed self-supervised learning with hierarchical masked pre-training, and Q. Xu *et al.* (2023) combined self-supervised learning with hybrid data augmentation strategies.

These methods mainly rely on attention mechanisms or domain adaptation strategies to address discrepancies in feature distributions. However, most of them still treat underwater acoustic spectrograms as generic visual images and perform holistic modeling, which may overlook the structured characteristics of ship-radiated noise in the frequency domain.

To address this problem, this paper proposes a sample-generation-based few-shot UATR framework with a knowledge-inspired spectral decomposition augmentation (SDA) strategy and a cross-feature guided recalibration module (CFM). The proposed framework combines structured spectrogram augmentation, multi-view representation learning, and task-adaptive fine-tuning to improve recognition robustness in complex marine environments.

1. A structured SDA strategy is proposed to generate complementary multi-view training samples in the spatial-frequency domain of spectrograms. A Siamese network with time-frequency attention and a CFM is further designed for robust spectrotemporal representation learning.
2. Cosine similarity is adopted to reduce sensitivity to global amplitude scaling, and multi-view joint learning is used to encourage shared discriminative representations across different views.
3. A task-adaptive fine-tuning mechanism based on pseudo-sample generation (PSG) is proposed to enrich the support-set distribution during testing and improve adaptation to unseen classes and cross-domain environments.

II. METHODS

A. SDA

In underwater acoustic environments, target signatures are often weak and easily corrupted by ambient noise and propagation effects, which makes few-shot recognition particularly challenging. To improve representation robustness under such conditions, we propose SDA, a structured augmentation strategy applied in the two-dimensional (2D) spatial-frequency domain of spectrograms. Unlike conventional spatiotemporal augmentations (e.g., flipping or cropping) that mainly increase sample diversity, SDA decomposes spectrogram patterns into coarse global structures and fine local texture variations, and then recombines

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them to construct complementary training views. Although this design is motivated by domain knowledge in underwater acoustics, it is not intended as a physically exact decomposition of target and noise components in the raw acoustic signal, but rather as a representation-level prior for structured augmentation.

Specifically, as illustrated in Fig. 1, given an input time-frequency spectrogram $X \in \mathbb{R}^{C\times H\times W}$, SDA first transforms the spectrogram into the 2D spatial-frequency domain via the fast Fourier transform (FFT), denoted as $\mathcal{F}(X) = \mathrm{FFT}(X)$, where the FFT is applied to the STFT spectrogram as a 2D feature matrix rather than to the raw time-domain waveform. Therefore, the coordinates in $\mathcal{F}(X)$ correspond to spatial frequencies of the spectrogram pattern, rather than the acoustic low- and high-frequency components of the original signal itself. To separate coarse spectrotemporal structures from fine local texture variations in the time-frequency representation, we introduce a binary rectangular mask M.

$$M(u,v) = \begin{cases} 1, & \text{if } |u-u_0| \le \dfrac{rH}{2} \text{ and } |v-v_0| \le \dfrac{rW}{2} \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

The retention ratio is empirically set to $r = 0.2$. After the 2D FFT, the region where $M = 1$ retains the central low-spatial-frequency components of the spectrogram pattern, which mainly describe coarse and smoothly varying spectrotemporal structures, whereas the region where $M = 0$ corresponds to high-spatial-frequency components that mainly capture fine local texture variations and noise-sensitive details.

1. Smooth augmented view (Y_{zero})

By suppressing the peripheral high-spatial-frequency components, SDA produces a smoothed view of the spectrogram pattern. Although this removes some fine local details, it preserves the main coarse spectrotemporal structure and reduces sensitivity to irregular texture variations in noisy conditions:

$$Y_{zero} = \mathcal{F}(X) \odot M, \quad (2)$$

where $\odot$ denotes element-wise multiplication.

2. Perturbed augmented view (Y_{randn})

This view preserves the central low-spatial-frequency structure while perturbing the peripheral high-spatial-frequency region with adaptive noise N. The goal is not to simulate a specific physical noise source directly, but to expose

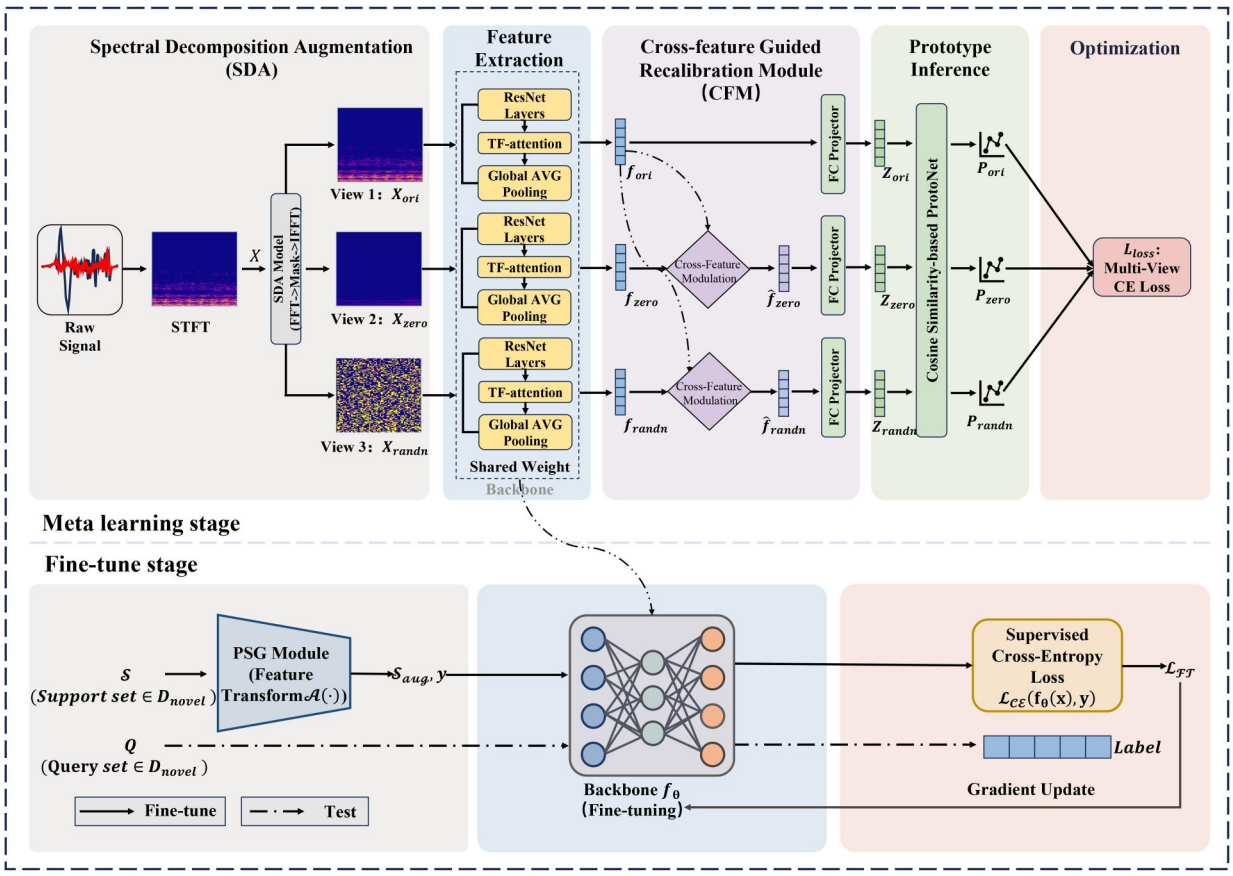


FIG. 1. Schematic diagram of the proposed SDA_CFM framework. The framework consists of two main stages: the upper part is the meta-learning stage based on SDA and CFM, aimed at learning robust spectrotemporal representations; the lower part is the task-adaptive fine-tuning stage based on PSG, designed for rapid adaptation to novel classes during testing.

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the model to plausible variations in fine spectrogram texture and thus improve robustness:

$$Y_{randn} = \mathcal{F}(X) \odot M + \mathcal{N} \odot (1 - M), \tag{3}$$

where the noise is defined as $\mathcal{N} \sim \mathcal{N}(0, \sigma^2)$, where σ is adaptively estimated from the standard deviation of the peripheral high-spatial-frequency coefficients of the transformed spectrogram. This keeps the perturbation magnitude consistent with the statistics of the original representation and avoids excessive synthetic artifacts.

Finally, the augmented representations $X_{aug} = \mathrm{IFFT}(Y_{aug})$ (where $aug \in \{zero, randn\}$) are reconstructed via the inverse FFT. It is important to note that this inverse FFT operation is performed on the transformed spectrogram features and is not intended to reconstruct physically meaningful time-domain waveforms, but rather to generate augmented feature representations for model training. Consequently, SDA constructs a three-stream input configuration comprising "original-smooth-perturbed" views. This design encourages the backbone network to learn representations that are robust to fine-grained texture perturbations and noise-induced variations in spectrogram patterns.

B. Siamese spectrotemporal feature extraction framework

To extract consistent, robust features from the multi-view inputs generated by SDA, we construct a Siamese network architecture that employs a weight-sharing mechanism. This framework is composed of a ResNet backbone integrated with an embedded time-frequency attention (TFA) module.

Let $\mathcal{S} = \{X_{ori}, X_{zero}, X_{randn}\}$ denote the set of input views, where $X_k \in \mathbb{R}^{C \times H \times W}$. We formalize the feature extractor as a mapping function $\Phi(\cdot; \theta)$, where θ represents the learnable parameters. During training, each view X_k in the set is fed into the shared network in parallel to generate feature embeddings:

$$Z_k = \Phi(X_k; \theta), \quad \forall X_k \in \mathcal{S}. \tag{4}$$

This design encourages the backbone to suppress perturbation-sensitive variations introduced by SDA and to focus on view-consistent structural representations.

For a given input X_k, preliminary encoding is performed via the convolutional layers of the ResNet backbone. Let $F_{in} \in \mathbb{R}^{C \times H' \times W'}$ denote the intermediate feature tensor output by the backbone. However, due to the local receptive fields of convolutional operations, the backbone struggles to capture long-range global dependencies in underwater acoustic signals. To mitigate this limitation, the TFA module is introduced to perform feature recalibration.

To capture complementary dependencies along the temporal and frequency dimensions, we design two parallel branches with symmetric topological structures to model global contextual information along both axes.

To obtain feature descriptors with global receptive fields, the input feature map is denoted by $F_{in} \in \mathbb{R}^{C \times H' \times W'}$, where C, H', and W' represent the channel number, frequency dimension, and time dimension, respectively. The frequency and temporal descriptors are computed as:

$$v^f = \frac{1}{W'} \sum_{w=1}^{W'} F_{in}[:, :, w], \quad v^t = \frac{1}{H'} \sum_{h=1}^{H'} F_{in}[:, h, :]. \tag{5}$$

$v^f \in \mathbb{R}^{C \times H' \times 1}$ is obtained by averaging along the time axis, which preserves frequency-wise information, while $v^t \in \mathbb{R}^{C \times 1 \times W'}$ is obtained by averaging along the frequency axis, which preserves temporal information.

Subsequently, to capture non-linear channel dependencies, we feed the descriptors into a multi-layer perceptron module with shared weights. The generation of the attention map A^k is unified as follows:

$$A^k = \sigma(W_2 \delta(W_1 v^k)), \quad k \in \{f, t\}, \tag{6}$$

where δ denotes the ReLU activation function, σ is the sigmoid function, and W_1, W_2 are the weight matrices of the bottleneck structure. Ultimately, A^f explicitly quantifies the importance of each frequency band, adaptively suppressing broadband background noise; conversely, A^t attends to time frames containing valid signals, ignoring pure background noise segments.

Finally, based on the learned complementary attention maps, we employ a broadcasting mechanism to perform element-wise weighting on the original features F_{in}, yielding the refined features F_{out}:

$$F_{out} = F_{in} \odot A^f \odot A^t. \tag{7}$$

For the perturbed input view, the network implicitly learns to suppress noise-sensitive high-spatial-frequency variations while preserving the dominant structural content of the spectrogram, thereby achieving a "soft denoising" effect at the feature level. By combining the structural representations learned by ResNet with the contextual recalibration provided by TFA, the model generates robust feature maps. Subsequently, GAP is employed to compress these maps into global feature vectors $f_{ori}, f_{zero}, f_{randn}$. Each vector encodes the aggregated information of its respective view across the entire spatial domain.

C. Cross-feature guided recalibration module

Building upon the global feature vectors extracted in the preceding section, we introduce the CFM module to attenuate noise inherent in the augmented views.

To bolster semantic consistency, the clean original feature, f_{ori}, serves as a guidance signal to guide the reweighting of the two augmented views, f_{zero} and f_{randn}, via a gating mechanism. Specifically, gating weights are derived from the element-wise interaction between the original and augmented features:

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$$\text{Weight} = \sigma(f_{ori} \odot f_{aug}), \tag{8}$$

where $\odot$ denotes the element-wise product and σ represents the sigmoid activation function. Subsequently, these weights are applied to the augmented features through a residual connection scheme:

$$\tilde{f}_{aug} = f_{aug} \odot (1 + \text{Weight}), \quad aug \in \{zero, randn\}. \tag{9}$$

From a signal processing perspective, underwater acoustic targets typically exhibit relatively stable and repeatable line-spectrum-related patterns, whose responses remain more consistent across different views. In contrast, environmental noise and interference are often non-stationary and usually show weaker correspondence across time–frequency locations. Based on this distinction, the element-wise interaction can be viewed as a soft consistency-aware weighting mechanism. Stable components that are consistently present across views receive stronger emphasis, while unstable components receive relatively weaker emphasis.

The proposed CFM module therefore acts as a data-driven soft emphasis mechanism, where the weighting coefficients are generated from cross-view consistency rather than predefined priors. This operation assigns stronger emphasis to stable spectral structures while giving relatively weaker emphasis to unstable or interference-sensitive components, thereby producing a spectral focus effect that encourages the network to concentrate on more discriminative regions.

To project these features into a low-dimensional metric space suitable for metric learning (depicted as the green module in Fig. 1), the original feature f_{ori} and the modulated augmented features $\tilde{f}_{aug}$ are passed through a fully connected projection layer. This layer compresses the high-dimensional features and performs a non-linear mapping to generate the final feature embeddings $Z = \{Z_{ori}, Z_{zero}, Z_{randn}\}$, which are subsequently fed into a prototypical network to compute the multi-stream classification loss.

D. Prototype-based meta-learning strategy

Building upon the robust feature vector set Z obtained previously, we employ a metric-based prototypical network to address the N-way K-shot few-shot classification problem.

During the training phase, each training task (referred to as an episode) comprises two disjoint subsets: a support Set ($\mathcal{S}$) and a query set ($\mathcal{Q}$). The Support Set, denoted as $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{N\times K}$, contains N classes with K samples per class and is utilized to construct class centroids. The query set, $\mathcal{Q} = \{(x_j, y_j)\}_{j=1}^{N\times Q_{cnt}}$, consists of the remaining samples from the aforementioned N classes and is employed for loss computation and gradient updates. For the kth class, the prototype $\mathbf{c}_k$ is defined as the empirical mean of all support samples within the feature space. For any given view $v \in \{ori, zero, randn\}$, the prototype is calculated as follows:

$$\mathbf{c}_k^v = \frac{1}{|\mathcal{S}_k|} \sum_{(x_i, y_i)\in\mathcal{S}_k} Z_v(x_i). \tag{10}$$

Regarding metric selection, underwater acoustic signals are significantly influenced by transmission distance and sea conditions, leading to severe, non-stationary fluctuations in magnitude. Traditional Euclidean distance is highly susceptible to interference from such unstructured amplitude variations. Consequently, we adopt cosine similarity as the metric to reduce the influence of signal intensity and emphasize structural information.

For a query sample x_q, the probability of it belonging to class k is given by a Softmax function with a learnable scaling factor α:

$$P_v(y = k|x_q) = \frac{\exp\left(\alpha \cos\left(Z_v(x_q), c_k^v\right)\right)}{\sum_{j=1}^{N} \exp\left(\alpha \cos\left(Z_v(x_q), c_j^v\right)\right)}, \tag{11}$$

where $\alpha \in \mathbb{R}^+$ is a learnable temperature scaling factor, $\cos(\cdot,\cdot)$ denotes cosine similarity, $c_k^v \in \mathbb{R}^d$ is the prototype of class k in view v, and N is the number of classes. This formulation encourages the model to emphasize relational structure in the embedding space while reducing sensitivity to global amplitude variations caused by propagation effects.

E. Multi-view shared-space learning objective

To simultaneously achieve fine-grained discrimination for underwater acoustic targets and environmental robustness against non-stationary conditions, we construct a joint optimization framework based on multi-view supervision with shared parameters. Distinct from traditional FSL approaches that perform metric learning solely within a single original domain, our method incorporates a hard parameter sharing mechanism. This mechanism encourages the backbone network to map features from the original, smoothed, and perturbed views into a shared discriminative prototype space.

To promote robustness in capturing line-spectrum features across the heterogeneous views generated by SDA, we minimize the joint negative log-likelihood loss of the three parallel branches on the query set:

$$\mathcal{L}_{total} = \frac{1}{|Q|} \sum_{(x_q, y_q)\in Q} [\mathcal{L}_{ce}(P_{ori}, y_q) + \mathcal{L}_{ce}(P_{zero}, y_q) + \mathcal{L}_{ce}(P_{randn}, y_q)], \tag{12}$$

where $\mathcal{L}_{ce}$ denotes the standard cross-entropy loss function, P_v represents the predicted probability distribution obtained after forward propagation for the input view $v \in \{ori, zero, randn\}$, and y_q denotes the ground truth class label. This formulation does not explicitly enforce cross-view consistency, but implicitly promotes representation alignment through shared parameters and joint supervision.

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TABLE I. Ship types and labels.

Label	0	1	2	3	4	5	6	7	8	9	10
Class	Dredgers	Fishing	Trawlers	Mussel	Tugboats	Motorboats	Pilot	Sailboats	Passenger	Ocean	Roro

F. Task-adaptive fine-tuning

To evaluate the model's generalization capability in unseen hydroacoustic environments, we test on a novel dataset ($\mathcal{D}_{novel}$) that is disjoint from the training categories. Following the standard episodic evaluation protocol, each test task is denoted as $T = \{S, Q\}$, where S and Q are the support and query sets, respectively. Unlike methods that rely only on static pre-trained features, we adopt an augmentation-based task-adaptive fine-tuning strategy. As illustrated in Fig. 1 (bottom), this strategy compensates for the sparsity of the support-set distribution by generating pseudo-samples, thereby improving adaptation to novel classes and domain shifts during testing.

Given the limited size of the support set S, we employ a PSG module to enrich the support-set distribution. For each support instance x_i, M pseudo-samples are generated through a stochastic, label-preserving perturbation function $\mathcal{A}(\cdot)$, forming the augmented support set S_{aug}:

$$S_{aug} = S \cup \left\{ (x_{i,m}, y_i) | x_{i,m} = \mathcal{A}(x_i), m = 1, \ldots, M \right\}. \quad (13)$$

$\mathcal{A}(\cdot)$ denotes a representation-level perturbation applied to spectrogram features during task-adaptive fine-tuning. Its role is to introduce simple stochastic variations while preserving class labels, thereby improving support-set diversity and reducing overfitting to the limited support instances. It is not intended as a physically exact acoustic transformation model.

Subsequently, gradient updates are performed on S_{aug}. In the inductive setting, the fine-tuning process relies exclusively on the labeled support set and does not access the query set. The fine-tuning loss $\mathcal{L}_{FT}$ is defined as:

$$\mathcal{L}_{FT}(\theta) = \frac{1}{|S_{aug}|} \sum_{(x,y) \in S_{aug}} \mathcal{L}_{ce}(f_\theta(x), y), \quad (14)$$

where $\mathcal{L}_{ce}$ denotes the supervised cross-entropy loss on the augmented support set. By minimizing $\mathcal{L}_{FT}$, the model parameters are updated from the initial state θ_{init} to the task-adapted parameters θ'. Finally, inference is performed on the query set Q as:

$$\hat{y} = \arg\max_c P(y = c | x_q; \theta'), \quad x_q \in Q.$$

To ensure rigorous evaluation, the model parameters are reset to θ_{init} after each task so that different episodes are evaluated independently. Details of the sample size and statistical reporting are provided in Sec. III.

III. EXPERIMENTAL SETUP

A. Datasets

This study constructs in-domain and cross-domain evaluation scenarios based on two public underwater acoustic datasets: ShipsEar and DanShip. All audio data are resampled to 16 kHz and segmented into 3-s fixed-length samples.

ShipsEar dataset (in-domain setting) (Santos-Domínguez *et al.*, 2016): We selected 11 classes of ship radiated noise data from this dataset, excluding background noise, as shown in Table I. To meet the requirements of FSL, we divided the categories into a base set and a novel set, which are strictly mutually exclusive at the class level. The base set is used for meta-training to learn general acoustic feature representations; the novel set is used for fine-tuning and evaluation during the meta-testing phase. To eliminate bias introduced by random partitioning, we used three different data splits, with the corresponding class assignments shown in Table II.

DanShip dataset (cross-domain setting) (Cui *et al.*, 2025): To verify the generalization ability of the model in heterogeneous environments, we introduced the DanShip dataset, which includes three target classes: fishing vessels, cargo ships, and motorboats. In this setting, the full ShipsEar dataset serves as the source domain for model training, while DanShip serves as the unseen target domain (novel set) for testing. This setting effectively simulates cross-domain scenarios in practical applications where there are significant acoustic differences between training data and deployment environments.

To evaluate the robustness of the model under low SNR conditions, we designed a non-stationary noise experimental protocol. Training phase: Natural ocean noise is randomly added to the base set samples, with SNR distributed uniformly over $[-15, 7]$ dB, to enhance the noise resistance of the feature extraction network through data augmentation. Testing phase: Four standardized test environments are constructed, including a clean environment without additional noise, and noisy environments with SNRs of –5, –10, and –15 dB, respectively.

TABLE II. Base and novel set split configuration.

Split	Base set	Novel set
1	0, 1, 3, 5, 8, 10	2, 4, 6, 7, 9
2	0, 1, 5, 7, 8, 10	2, 3, 4, 6, 9
3	1, 3, 5, 8, 9, 10	0, 2, 4, 6, 7

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B. Experimental details

1. Model input

STFT-spectrograms generated with 128 filters, a window length of 40 ms, and an overlap of 20 ms. The classification head employs a cosine similarity-based metric module to enhance feature space discriminability.

2. Training and fine-tuning

The Adam optimizer is utilized for parameter updates throughout the process. During the testing phase, task-adaptive fine-tuning is performed; specifically, the PSG module generates 75 pseudo-samples to augment the support set distribution, with the fine-tuning learning rate set to 0.005.

3. Task construction and stratified sampling

To improve the robustness of the meta-learning process and reduce reliance on the original data distribution, we adopt a stratified episodic sampling strategy to construct both training and evaluation tasks. In each episode, N classes are first sampled from the class set with equal probability, ensuring that underrepresented and majority classes have the same chance of being selected. Then, K support samples and a fixed number of query samples are independently and uniformly drawn from the candidate pool of each selected class, with no overlap between the two sets.

4. Evaluation protocol

We conduct extensive experiments on the ShipsEar and DanShip datasets. Following the data splits defined in Table II, we randomly sample 200 few-shot tasks (episodes) for testing and report the mean accuracy $\pm$ 95% confidence interval. All reported results are further averaged over three independent dataset splits. The experiments are implemented in PyTorch on a single NVIDIA vGPU with 32 GB VRAM.

5. Fairness of baseline comparison

All compared methods are evaluated under the same dataset splits, the same STFT-based input representation, and the same evaluation metrics. To preserve fidelity to the original methods, method-specific inference or adaptation mechanisms are retained rather than being forcibly unified. For methods with test-time fine-tuning or adaptation, evaluation follows their corresponding adaptation procedure; for methods without adaptation, evaluation follows their standard inference procedure. Therefore, the comparison is conducted under a unified data and reporting protocol while preserving the original characteristics of each method.

C. Evaluation metrics

To comprehensively quantify the model's recognition performance in non-stationary underwater acoustic environments, we select accuracy (ACC), F1-score, and area under the ROC curve (AUC) as core evaluation indicators.

ACC is the most straightforward performance metric, defined as the ratio of correctly classified samples to the total number of samples. For a K-class classification task, it is computed as:

$$\mathrm{ACC} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i = y_i], \tag{15}$$

where N is the total number of test samples, y_i is the ground-truth label of the i-th sample, $\hat{y}_i$ is the predicted label, and $\mathbf{1}[\cdot]$ is the indicator function.

1. F1-score

Since class imbalance may exist in underwater acoustic datasets, accuracy alone can be misleading. We therefore adopt the macro-averaged F1-score as a complementary metric. For each class $k \in \{1, \ldots, K\}$, the per-class precision and recall are defined as:

$$P_k = \frac{TP_k}{TP_k + FP_k}, \quad R_k = \frac{TP_k}{TP_k + FN_k},$$
$$\mathrm{F1}_k = \frac{2 \cdot P_k \cdot R_k}{P_k + R_k}. \tag{16}$$

The macro-averaged F1-score treats all classes equally and is computed as:

$$\mathrm{F1}_{\mathrm{macro}} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{F1}_k, \tag{17}$$

where TP_k, FP_k, and FN_k denote the number of true positives, false positives, and false negatives for class k, respectively.

2. AUC

To evaluate the discriminative ability of the classifier in the multi-class setting, we employ the macro-averaged AUC under the One-vs-Rest scheme. For each class k, an ROC curve is constructed by considering class k as the positive class and all remaining classes as the negative class. The corresponding area under this curve is denoted as AUC_k. The final AUC is then obtained by macro-averaging over all classes:

$$\mathrm{AUC}_{\mathrm{macro}} = \frac{1}{K}\sum_{k=1}^{K} \mathrm{AUC}_k. \tag{18}$$

3. Separability score (S)

To further quantify the discriminative power of the learned representations, we introduce S as a feature space characterization metric. The score is computed in the original high-dimensional feature space before applying any dimensionality reduction techniques such as t-SNE or UMAP.

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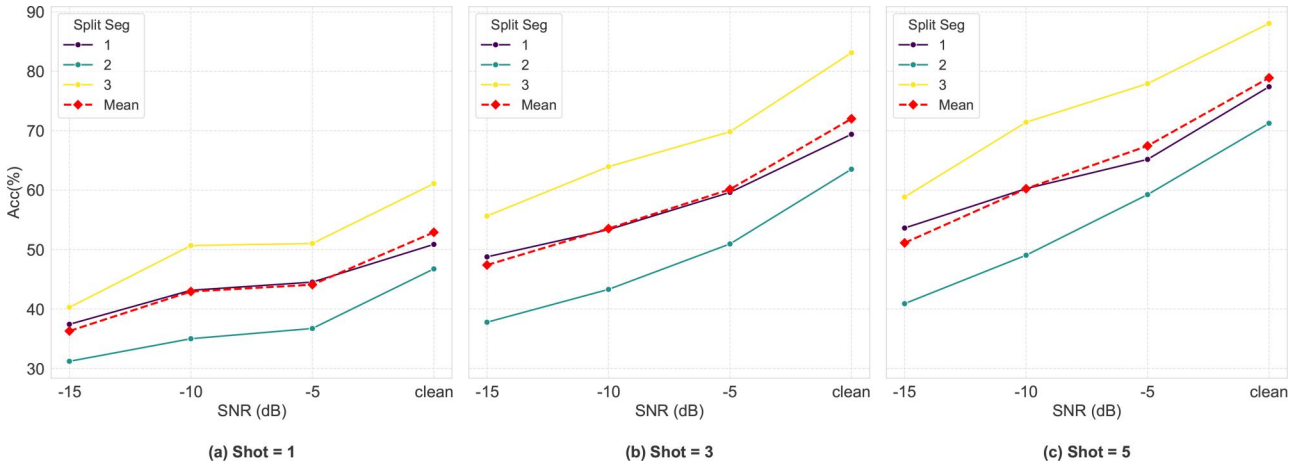


FIG. 2. Performance comparison of different data split methods under different SNRs and shots.

This metric evaluates the ratio between inter-class dispersion and intra-class compactness, and is related to the classical Fisher discriminant criterion. Specifically, let C denote the number of classes, $\boldsymbol{\mu}_i$ the centroid of class i, and $\boldsymbol{\mu}_g$ the global centroid of all samples. The separability score is defined as:

$$S = \frac{D_{\text{inter}}}{D_{\text{intra}}} = \frac{\frac{1}{C}\sum_{i=1}^{C} ||\boldsymbol{\mu}_i - \boldsymbol{\mu}_g||^2}{\frac{1}{C}\sum_{i=1}^{C}\left(\frac{1}{n_i}\sum_{\boldsymbol{x}\in X_i} ||\boldsymbol{x} - \boldsymbol{\mu}_i||^2\right)}, \tag{19}$$

where X_i and n_i denote the sample set and sample size of class i, respectively. A higher S value indicates better class separability, with larger inter-class dispersion and more compact intra-class distributions. This metric provides a quantitative complement to the qualitative visualization results.

IV. EXPERIMENTAL ANALYSIS

A. Analysis of results under different data splits

As shown in Fig. 2, there is a clear performance gap among the three data splits. Split 3 consistently achieves the highest accuracy, whereas split 2 performs the worst, with the most noticeable drop under the –15 dB condition. This trend is consistent with the t-SNE visualization in Fig. 3, where split 3 exhibits better cluster separation (score: 0.648), while split 2 shows more severe overlap among classes (score: 0.482).

Figure 4 further illustrates the class-wise prediction behavior through confusion matrices. In split 2, class C_6 has the lowest recognition rate among all categories across the three splits, with a diagonal value of only 0.59. Its misclassifications are distributed across several categories, including C_2 (0.11) and C_3 (0.11), indicating substantial confusion with similar classes. A noticeable bidirectional confusion is also observed between C_5 and C_6. In contrast, the confusion matrix of split 3 shows consistently strong class separability, with diagonal values ranging from 0.79 to 0.85 and most off diagonal values remaining around 0.03–0.06.

The performance degradation in split 2 can be explained by its class composition. In this split, several highly similar categories, including mussel (C_3), pilot (C_6), and ocean (C_9), are all assigned to the novel set. In particular, the cosine similarity between C_3 and C_9 reaches 0.9678, which makes recognition more difficult under the few-shot setting. By contrast, in split 3, these confusing categories are included in

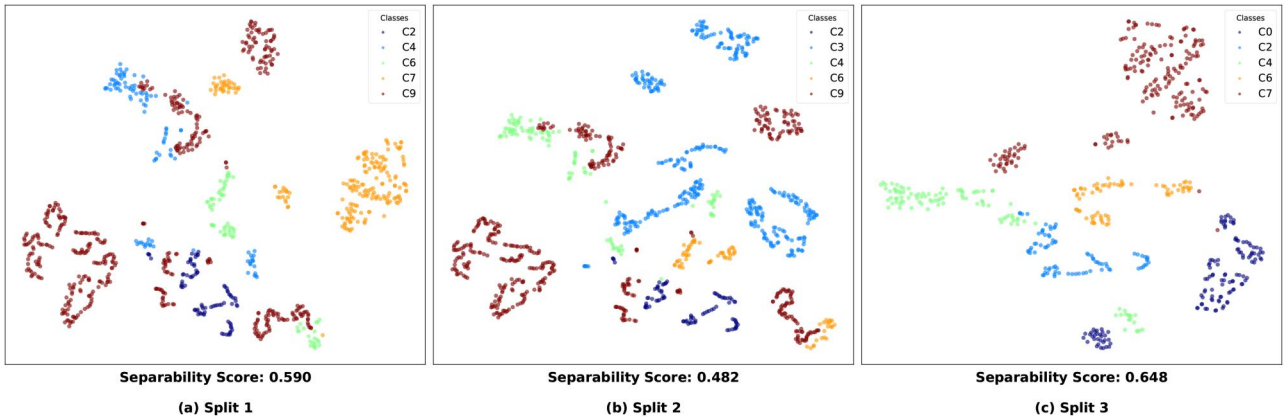


FIG. 3. t-SNE visualization and clustering quality under different data splits.

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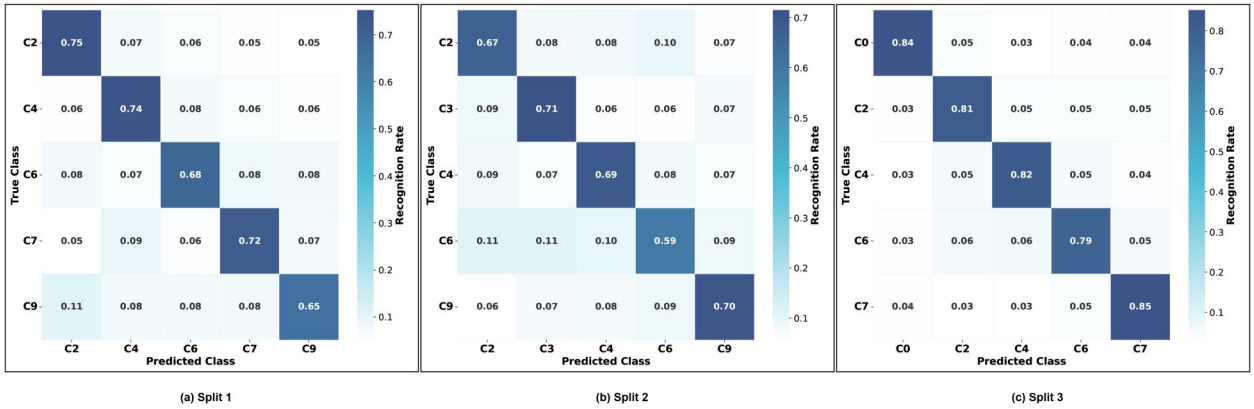


FIG. 4. Confusion matrices for the three data splits under the 5-shot setting.

the base set, allowing the model to learn more discriminative features during pre-training. Since the difficulty varies substantially across splits, we report the average performance over all three splits to ensure a fair and robust evaluation.

B. Comparison with state-of-the-art UATR models

To comprehensively evaluate the performance of the proposed SDA_CFM method on underwater acoustic target recognition, several representative few-shot learning approaches are selected as baselines. These include metric-based methods (RelationNet, MatchingNet, and ProtoNet), as well as advanced methods designed for few-shot scenarios (RFS, LabelHallu, KT, and DAATF). Experiments on the in-domain dataset are conducted from two perspectives: recognition performance on novel classes under clean conditions, and robustness under different SNRs.

For in-domain evaluation on novel classes, experiments are conducted under 1-shot, 3-shot, and 5-shot settings. ACC, F1-score, and AUC are adopted as evaluation metrics. All results are reported as the mean over 200 episodes with three random splits, along with 95% confidence intervals, as summarized in Table III.

Overall, SDA_CFM achieves the best performance across all evaluation metrics. Under the 1-shot setting, SDA_CFM obtains an accuracy of 53.82% ± 1.37%, an F1-score of 53.22%, and an AUC of 79.71%. Compared with RelationNet, the improvements are 20.51%, 22.13%, and 16.24%, respectively. Compared with the second-best method DAATF, SDA_CFM improves the accuracy by 2.96%, while reducing the confidence interval from ±1.95% to ±1.37%, indicating more stable predictions under limited support samples.

Under the 3-shot setting, SDA_CFM achieves an accuracy of 71.03% ± 0.91%, an F1-score of 70.64%, and an AUC of 90.07%, outperforming DAATF by 2.97% in accuracy with a narrower confidence interval. Under the 5-shot setting, the accuracy further increases to 78.54% ± 0.75%, with an F1-score of 78.24% and an AUC of 93.33%. The accuracy gain over DAATF reaches 3.87%, and the confidence interval is reduced from ±1.55% to ±0.75%. These results indicate that SDA_CFM provides consistent improvements in both recognition performance and prediction stability in in-domain few-shot scenarios.

To further evaluate robustness in low-SNR environments, we compare all methods under different noise levels in the 5-shot setting. The results are summarized in Table IV. As the SNR decreases, all methods show performance degradation, but the extent of degradation varies.

At SNR = –5 dB, SDA_CFM achieves an accuracy of 66.86% ± 0.84%, exceeding DAATF by 4.09%, while reducing the confidence interval from ±1.63% to ±0.84%. At SNR = –10 dB, SDA_CFM maintains an accuracy of 60.05% ± 0.85%, outperforming DAATF by 5.17%. Under the more challenging condition of SNR = –15 dB, SDA_CFM still

TABLE III. Performance comparison of few-shot classification methods on the clean novel set of ShipsEar dataset. Boldface values indicate the best-performing result for each metric in the corresponding comparison.

	1-shot			3-shot			5-shot		
Model	ACC ± 95% CI	F1 (%)	AUC (%)	ACC ± 95% CI	F1 (%)	AUC (%)	ACC ± 95% CI	F1 (%)	AUC (%)
RelationNet	33.31 ± 7.71	31.09	63.47	38.53 ± 7.32	36.07	69.83	36.65 ± 6.63	32.20	70.32
MatchingNet	36.59 ± 8.36	34.47	64.11	44.44 ± 7.03	39.96	72.87	44.53 ± 6.25	40.66	73.89
ProtoNet	39.91 ± 9.16	38.25	69.62	50.41 ± 8.01	49.34	78.58	54.81 ± 6.87	53.77	81.80
RFS	46.29 ± 0.76	44.57	74.54	61.99 ± 0.57	61.42	84.81	66.82 ± 0.52	66.43	87.45
LabelHallu	47.01 ± 2.78	45.29	75.69	64.86 ± 2.03	64.26	87.17	71.26 ± 1.87	70.68	90.28
KT	48.81 ± 1.76	48.13	74.10	66.61 ± 1.55	65.83	85.94	73.87 ± 1.48	73.12	89.93
DAATF	50.86 ± 1.95	50.31	76.85	68.06 ± 1.65	68.22	89.34	74.67 ± 1.55	75.95	92.61
SDA-CFM (ours)	**53.82 ± 1.37**	**53.22**	**79.71**	**71.03 ± 0.91**	**70.64**	**90.07**	**78.54 ± 0.75**	**78.24**	**93.33**

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TABLE IV. Performance comparison of few-shot classification methods on the noisy novel set of ShipsEar dataset. Boldface values indicate the best-performing result for each metric in the corresponding comparison.

	ACC ± 95% CI		
Model	−15 dB	−10 dB	−5 dB
RelationNet	26.88 ± 5.62	33.96 ± 5.43	37.17 ± 5.57
MatchingNet	29.42 ± 4.97	32.59 ± 5.07	36.78 ± 4.92
ProtoNet	28.57 ± 5.65	37.76 ± 6.85	43.21 ± 7.22
RFS	40.75 ± 0.52	50.79 ± 0.55	56.64 ± 0.55
LabelHallu	40.07 ± 1.98	49.80 ± 1.95	56.57 ± 2.12
KT	44.28 ± 1.50	55.25 ± 1.51	61.64 ± 1.49
DAATF	43.64 ± 1.63	54.88 ± 1.66	62.77 ± 1.63
SDA-CFM (ours)	**51.00 ± 0.87**	**60.05 ± 0.85**	**66.86 ± 0.84**

achieves 51.00% ± 0.87%, whereas RelationNet obtains 26.88% ± 5.62%, resulting in a gap of over 24% points.

It is also observed that metric-based methods (RelationNet, MatchingNet, and ProtoNet) exhibit relatively wide confidence intervals under low-SNR conditions (ranging from ±4.92% to ±7.22%), indicating less stable predictions. In contrast, SDA_CFM maintains confidence intervals within ±0.87% across all SNR levels, reflecting more consistent performance. These results suggest that SDA_CFM is more effective at preserving discriminative information under noise interference, leading to improved robustness in low-SNR few-shot recognition tasks.

C. Cross-dataset robustness analysis on DanShip

To evaluate the generalization ability of SDA_CFM under cross-domain feature shifts, we use the DanShip dataset as the target domain, which differs substantially from ShipsEar in acquisition conditions. No DanShip data are used during pre-training, ensuring a fair cross-domain evaluation.

The cross-domain results under clean conditions are reported in Table V. SDA_CFM achieves the best accuracy under all support settings. Specifically, in the 1-shot setting, SDA_CFM reaches 62.52% ± 1.63%, outperforming the second-best method, DAATF (58.40% ± 1.09%), by 4.12%. In the 3-shot setting, the accuracy increases to 74.84% ± 1.19%, exceeding DAATF by 10.77% and KT (67.73% ± 1.64%) by 7.11%. In the 5-shot setting, SDA_CFM further improves to 80.29% ± 0.99%, which is 5.16% higher than DAATF.

It is also worth noting that the metric-based methods, including RelationNet, MatchingNet, and ProtoNet, exhibit relatively wide confidence intervals in the cross-domain setting, indicating larger performance fluctuations. By contrast, SDA_CFM maintains comparatively narrow confidence intervals across all support settings, suggesting more stable transfer performance. These results demonstrate that the proposed method can better alleviate domain discrepancy and learn more consistent representations across different acoustic environments.

To further evaluate robustness under more challenging conditions, we conduct additional cross-domain experiments on DanShip with injected noise at SNR = −5, −10, and −15 dB. This setting introduces domain shift and noise interference simultaneously. The results are summarized in Table VI.

As expected, all methods show performance degradation compared with the clean condition. Nevertheless, SDA_CFM achieves the highest accuracy at all tested SNR levels. At SNR = −5 dB, SDA_CFM achieves 67.59% ± 1.12%, outperforming DAATF (65.67% ± 1.76%) by 1.92%. At SNR = −10 dB, it reaches 62.54% ± 1.18%, which is 1.67% higher than DAATF (60.87% ± 1.82%). At SNR = −15 dB, SDA_CFM achieves 55.84% ± 1.15%, surpassing RFS (55.05% ± 0.72%) and DAATF (54.22% ± 1.84%) by 0.79% and 1.62%, respectively.

In addition, SDA_CFM maintains relatively narrow confidence intervals across all noisy cross-domain settings, indicating more stable predictive behavior under combined domain shift and low-SNR interference. Overall, these results verify that SDA_CFM preserves discriminative information more effectively in challenging cross-domain environments.

D. Ablation experiments

To comprehensively evaluate the effectiveness of each key module in the proposed method, this section conducts systematic ablation experiments on the ShipsEar dataset. All

TABLE V. Performance comparison of few-shot classification methods on the clean novel set of DanShip dataset. Boldface values indicate the best-performing result for each metric in the corresponding comparison.

	1-shot	3-shot	5-shot
Model	ACC ± 95% CI	ACC ± 95% CI	ACC ± 95% CI
RelationNet	49.95±9.94	51.05±9.86	54.99±7.08
MatchingNet	53.29±11.93	57.58±7.66	66.54±7.07
ProtoNet	51.96±13.64	56.87±8.59	66.41±7.98
RFS	53.03±0.94	64.01±0.75	71.48±0.65
LabelHallu	52.89±1.70	63.14±1.34	70.76±1.14
KT	54.47±1.00	67.73±1.64	73.32±1.53
DAATF	58.40±1.09	64.07±1.80	75.13±1.64
SDA-CFM (ours)	**62.52±1.63**	**74.84±1.19**	**80.29±0.99**

TABLE VI. Performance comparison of few-shot classification methods on the noisy novel set of DanShip dataset (5-shot). Boldface values indicate the best-performing result for each metric in the corresponding comparison.

	ACC ± 95% CI		
Model	−15 dB	−10 dB	−5 dB
RelationNet	38.98±7.19	50.36±7.96	59.58±8.37
MatchingNet	40.09±7.63	44.90±7.56	49.76±9.09
ProtoNet	41.44±8.19	48.83±8.25	58.31±8.94
RFS	55.05±0.72	58.71±0.71	64.71±0.67
LabelHallu	54.93±1.28	57.49±1.27	64.56±1.18
KT	54.12±1.70	60.25±1.64	64.99±1.61
DAATF	54.22±1.84	60.87±1.82	65.67±1.76
SDA-CFM (ours)	**55.84±1.15**	**62.54±1.18**	**67.59±1.12**

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TABLE VII. Ablation study on key components [5-shot accuracy (%) ± 95% CI]. Boldface values indicate the best-performing result for each metric in the corresponding comparison.

Model	SDA	TFA	CFM	PSG	−15 dB	−10 dB	−5 dB	Clean
Baseline	×	×	×	✓	37.28±0.881	50.81±0.941	58.89±0.903	76.99±0.771
Configuration 2	×	✓	×	✓	37.24±0.887	50.76±0.929	58.89±0.914	76.94±0.775
Configuration 3	✓	×	×	✓	49.65±0.868	59.42±0.771	64.83±0.866	76.19±0.797
Configuration 4	×	×	✓	✓	45.69±0.883	56.69±0.852	63.98±0.825	77.08±0.713
Configuration 5	✓	✓	×	✓	49.95±0.866	59.28±0.804	65.23±0.849	77.70±0.757
Configuration 6	✓	×	✓	✓	50.64±0.885	59.27±0.796	65.00±0.839	78.56±0.748
Configuration 7	×	✓	✓	✓	37.03±0.885	50.56±0.910	58.82±0.912	76.96±0.780
Configuration 8	✓	✓	✓	×	39.43±0.856	42.83±0.851	44.39±0.897	52.01±0.907
SDA-CFM	✓	✓	✓	✓	**51.00 ± 0.869**	**60.05 ± 0.848**	**66.86 ± 0.845**	**78.54 ± 0.750**

experiments adopt a 5-shot setting and are tested under four SNR conditions (clean, −5 dB, −10 dB, and −15 dB) to validate the robustness contribution of each module in complex noisy environments.

1. Ablation study on model components

To evaluate the contribution of each module, we conduct systematic ablation experiments under the 5-shot setting, with results summarized in Table VII.

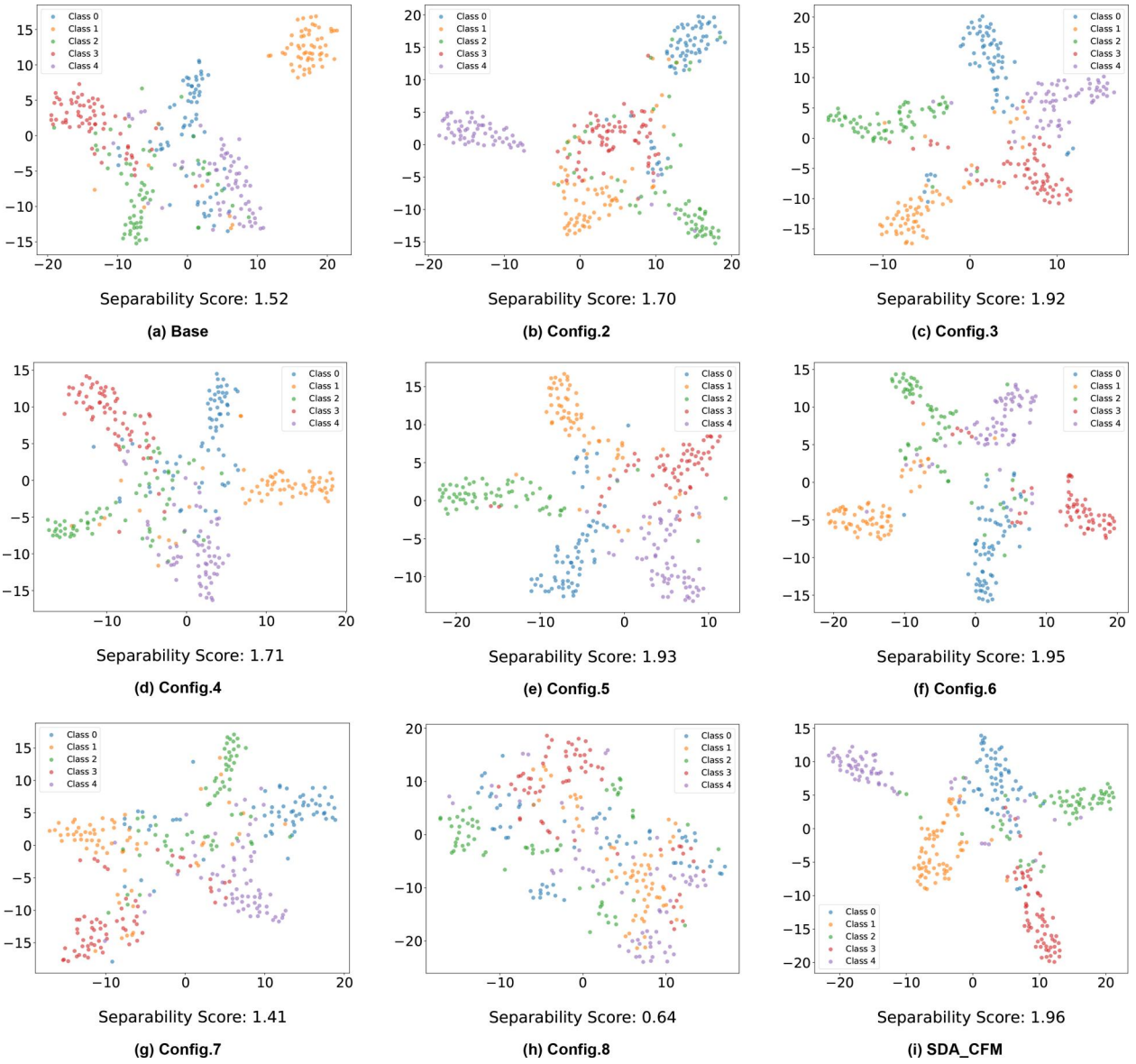


FIG. 5. t-SNE visualization of feature distributions under different configurations.

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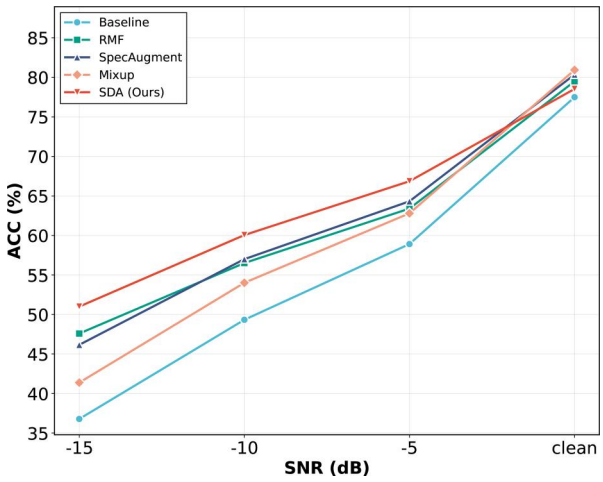


FIG. 6. Comparison with different data augmentation methods.

The baseline model using only PSG achieves 76.99% accuracy under clean conditions, but degrades to 37.28% at SNR = −15 dB. Introducing SDA (configuration 3) raises low-SNR accuracy to 49.65% (+12.37 pp) with negligible clean-condition loss (76.19%), confirming SDA's effectiveness in noisy scenarios. Regarding CFM, applying it directly to original features (configuration 4) or combining it with TFA without SDA (configuration 7) yields only marginal gains. However, when built upon SDA-enhanced features (configuration 6), CFM achieves consistent improvements across most conditions, indicating that its effectiveness relies on high-quality input representations provided by SDA. Removing PSG (configuration 8) causes a substantial drop in clean-condition accuracy from 78.54% to 52.01%, with similar degradation across noisy conditions. This is attributed to insufficient support set diversity, which undermines prototype estimation and prevents SDA, TFA, and CFM from being fully utilized—highlighting PSG's critical role in stabilizing metric learning.

The t-SNE visualizations in Fig. 5 further corroborate these findings. The baseline shows scattered distributions with blurred boundaries, while configuration 6 produces clearer clustering. Configuration 8 exhibits severe class overlap despite retaining all feature modules, and the complete SDA_CFM model achieves the most compact and well-separated feature distribution.

Overall, SDA constructs noise-robust complementary views, CFM filters redundancy using the original signal as reference, and PSG calibrates task distribution through support-set augmentation. Their synergy enables SDA_CFM to achieve robust recognition across varying SNR conditions.

2. Comparison of data augmentation strategies

We compare five data augmentation strategies, including no augmentation, random frequency masking, SpecAugment, Mixup, and the proposed SDA, as shown in Fig. 6. Figure 7 illustrates the spectrograms produced by each method on the same input sample.

Under noisy conditions, SDA consistently outperforms all baselines. At SNR = −15 dB, SDA achieves 51.00%, surpassing the no-augmentation baseline (36.78%), random frequency masking (47.59%), SpecAugment (46.16%), and

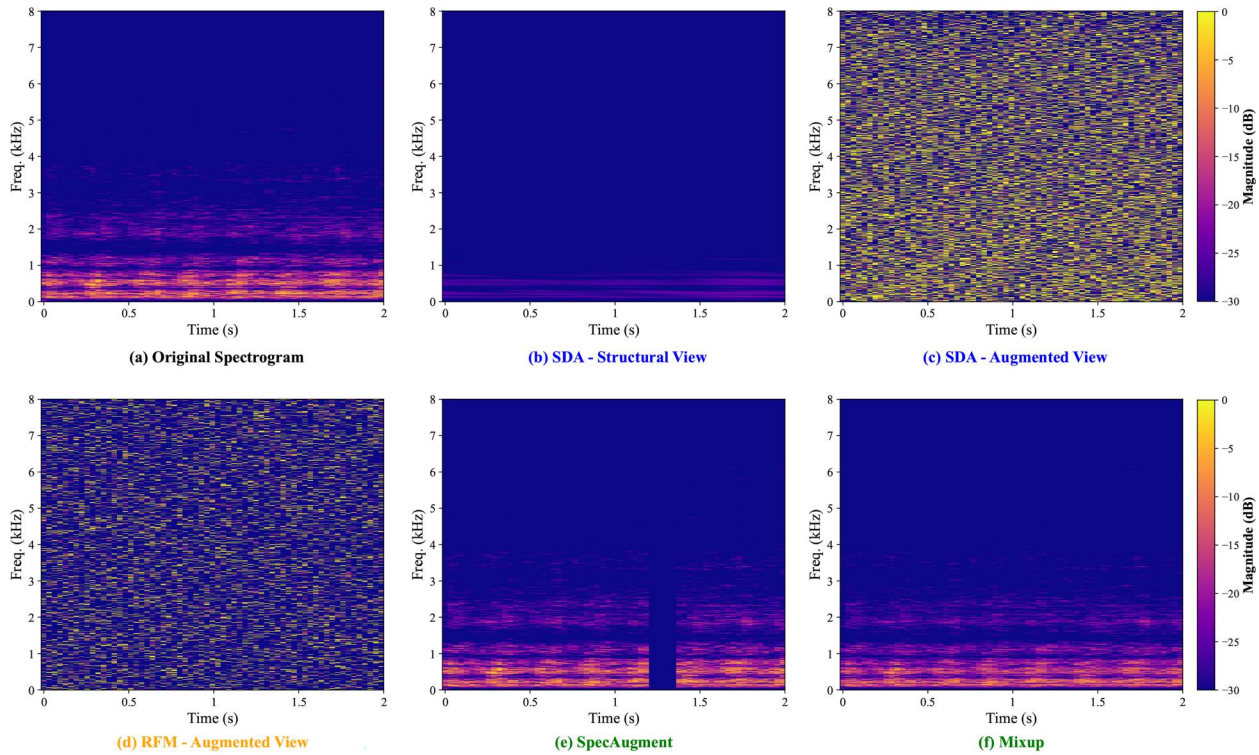


FIG. 7. Spectrograms of different data augmentation methods.

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Mixup (41.38%). This advantage is maintained at –10 dB (60.05%) and –5 dB (66.86%). The ∼4 percentage-point gap between random frequency masking and SDA, despite sharing the same technical framework, highlights the benefit of the proposed central-mask design over random masking positions.

Under clean conditions, SpecAugment (80.36%) and Mixup (80.94%) slightly exceed SDA (78.54%), as these general-purpose methods prioritize sample diversity rather than noise robustness. Given that real marine environments typically operate at low SNR, SDA's consistent advantage in noisy conditions holds greater practical value.

3. Ablation study on multi-view configurations

To validate the effectiveness of the multi-view collaborative architecture, we conduct ablation experiments across different SNR conditions, with results shown in Fig. 8. The single-view baseline achieves 77.50% accuracy under clean conditions, but drops to 36.78% at –15 dB, reflecting the limitations of a single feature stream in noisy environments.

Naively introducing a smooth or perturbed view (+smooth /+perturbed) yields no improvement; instead, it causes notable degradation across all SNR levels (e.g., +smooth drops to 23.92% at –15 dB). This is because, without an effective feature alignment mechanism, the added views introduce non-discriminative heterogeneous noise that disrupts the feature space distribution and confuses prototype-based inference.

In contrast, Multi-View (ours) achieves the best performance across all conditions, with a 14.22% improvement over the baseline at –15 dB. This gain is attributed to the CFM module, which uses the original view as a reference and dynamically calibrates augmented views via an asymmetric gating mechanism, rather than naively concatenating multiple streams. Combined with joint training, the model effectively exploits view diversity while maintaining structural alignment in the embedding space, resulting in consistently robust recognition.

TABLE VIII. Performance comparison under different SDA masking ratios r [5-shot accuracy (%) ± 95% CI]. Boldface values indicate the best-performing result for each metric in the corresponding comparison.

r	SNR = –15	SNR = –10	SNR = –5	SNR = clean
0.1	47.48±0.894	58.60±0.848	65.28±0.833	78.30±0.729
0.2	**51.00 ± 0.869**	60.05±0.848	**66.86 ± 0.845**	**78.54 ± 0.750**
0.4	50.76±0.876	60.01±0.854	65.19±0.855	77.66±0.761
0.6	50.73±0.873	**60.45 ± 0.819**	65.08±0.819	77.65±0.778
0.8	50.40±0.872	59.92±0.832	65.51±0.849	78.31±0.745

4. Analysis of SDA and masking design

a. Masking ratio sensitivity. The effect of the masking ratio r in the SDA module is analyzed under different noise conditions. As shown in Table VIII, $r = 0.2$ achieves the best performance in most cases, reaching 51.00%, 60.05%, 66.86%, and 78.54% at SNR levels of –15, –10, –5 dB, and clean conditions, respectively. Although $r = 0.6$ yields slightly higher accuracy at SNR = –10 dB, the margin is within 0.4%, indicating low sensitivity to this parameter. The consistency of confidence intervals across settings further confirms robustness with respect to r. Accordingly, $r = 0.2$ is adopted as the default in subsequent experiments.

b. Comparison of masking strategies. The proposed central-mask design is compared with random, circular, high-frequency, and band-prior-based masking strategies. As shown in Fig. 9, the proposed method achieves the best overall performance with an average accuracy of 64.11%. At SNR = –15 dB, it reaches 51.00%, outperforming the second-best method (band_prior) by 1.03%, with the largest margin of 2.79% observed at SNR = –15 dB. Random masking performs competitively under clean conditions but degrades in noisy scenarios, while high-frequency masking consistently underperforms, suggesting that high-frequency components are less reliable under severe noise. These results demonstrate that the proposed strategy more effectively preserves discriminative information while suppressing noise sensitivity.

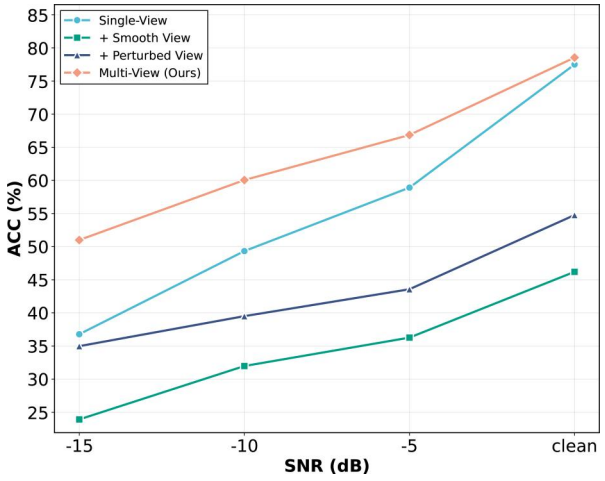


FIG. 8. Comparison of ablation results under different view configurations.

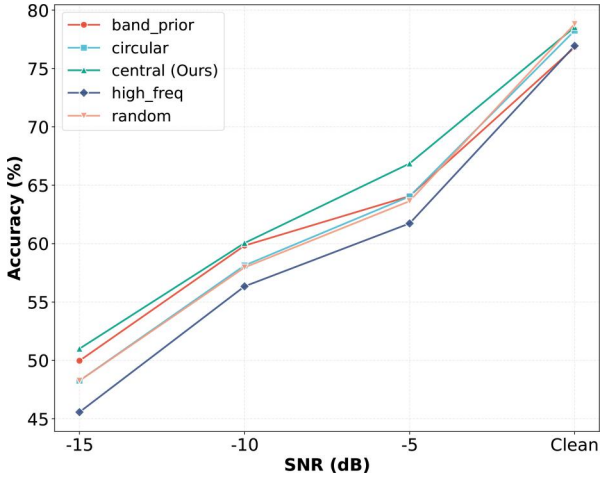


FIG. 9. Effectiveness of masking strategy.

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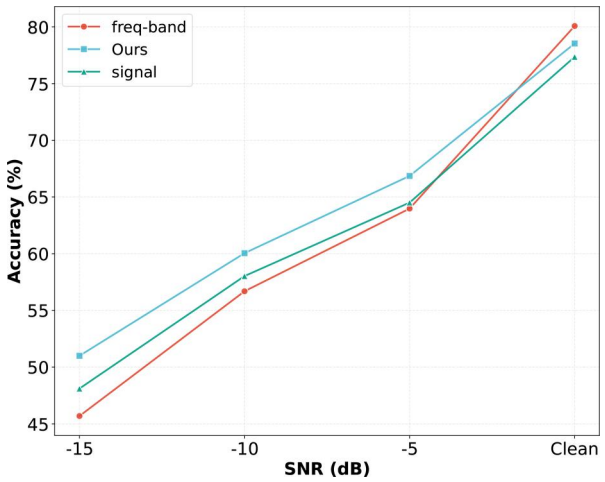


FIG. 10. Effectiveness of SDA under different augmentation strategies.

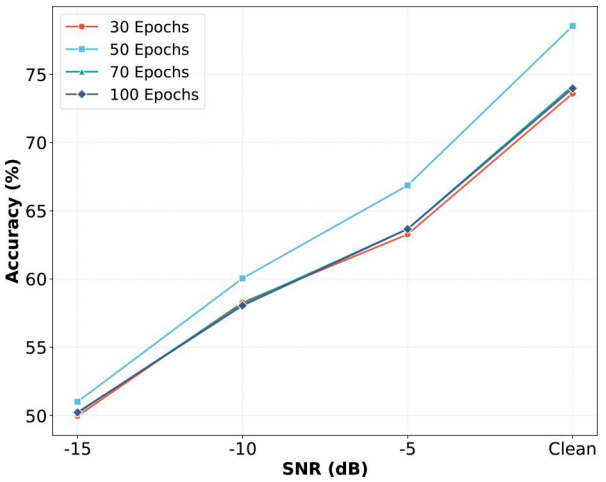


FIG. 11. Sensitivity analysis of the number of fine-tuning epochs.

c. Effectiveness of SDA. SDA is compared with representative signal-domain and frequency-band-domain augmentation strategies, as shown in Fig. 10. SDA consistently outperforms both alternatives, improving classification accuracy by 2.89% and 5.31% over signal-domain and frequency-band-domain methods at SNR = –15 dB, respectively. Signal-domain perturbations risk distorting the temporal structure of acoustic signals, while fixed frequency-band filtering lacks adaptability in complex noise environments. In contrast, SDA applies structured spectral perturbations via a corner-mask, selectively suppressing specific spectral regions to encourage the model to learn robust representations from partially degraded inputs.

5. Ablation studies on key hyperparameters

a. Effect of pseudo-samples. The effect of the number of synthetic samples N on model generalization under the 5-shot setting is examined, with results shown in Table IX. Increasing N from 25 to 75 yields consistent accuracy improvements, particularly under strong noise conditions, with accuracy at SNR = –15 dB rising from 50.64% to 51.00%. This is consistent with the meta-learning interpretation that greater synthetic diversity mitigates overfitting caused by limited labeled samples. Further increasing N to 100 introduces diminishing returns and slight degradation in certain noise regimes, likely due to distribution mismatch between pseudo-samples and original features. Balancing accuracy and computational cost, $N = 75$ is adopted as the default configuration.

TABLE IX. Performance comparison under different numbers of pseudo-samples N [5-shot accuracy (%) ± 95% CI]. Boldface values indicate the best-performing result for each metric in the corresponding comparison.

N_{Pseudo}	SNR = –15	SNR = –10	SNR = –5	SNR = clean
25	50.64±0.889	58.29±0.891	63.55±0.877	75.35±0.817
50	50.70±0.863	59.46±0.872	65.19±0.880	77.06±0.786
75	**51.00 ± 0.869**	60.05±0.848	**66.86 ± 0.845**	**78.54 ± 0.750**
100	50.89±0.848	**60.49 ± 0.812**	65.34±0.892	78.10±0.767

b. Effect of fine-tuning epochs. The effect of fine-tuning epochs on accuracy and generalization is investigated, as shown in Fig. 11. Model performance follows a non-monotonic trend: accuracy increases from 49.96% to 51.00% at SNR = –15 dB as epochs grow from 30 to 50, then degrades consistently beyond 70 epochs. The early improvement reflects more thorough extraction of discriminative information from synthetic samples, while the later decline is attributed to overfitting to the synthetic distribution. Accordingly, 50 fine-tuning epochs are adopted as the optimal setting.

c. Effect of fine-tuning learning rate. The sensitivity of model performance to the fine-tuning learning rate (LR) is analyzed, as shown in Fig. 12. A small LR (e.g., 0.0001) overly constrains gradient updates, limiting task-specific adaptation, while a large LR (e.g., 0.05) causes parameter drift and catastrophic forgetting. $LR = 0.005$ yields the most stable performance across all noise levels, achieving a peak accuracy of 51.00% at low SNR, effectively balancing prior knowledge retention and target-domain adaptation.

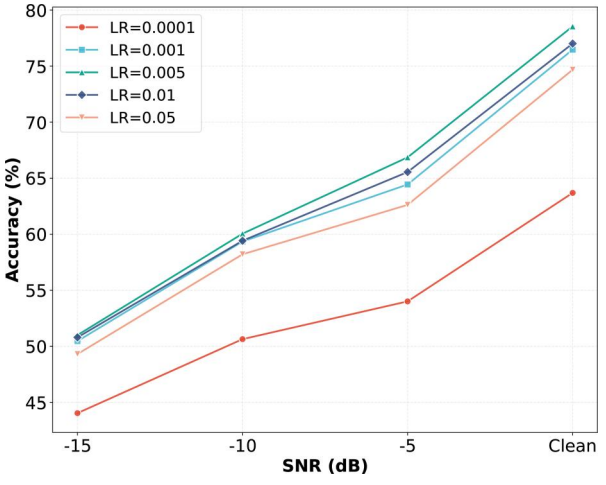


FIG. 12. Sensitivity analysis of the fine-tuning learning rate.

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TABLE X. Comparison of computational complexity and inference latency among different models. Boldface values indicate the best-performing result for each metric in the corresponding comparison.

Model	Parameters (M)	FLOPs (G)	Inference latency (ms)
RelationNet	0.3	92.14	39.77
MatchingNet	0.04	8.07	3.39
ProtoNet	0.04	8.07	3.37
RFS	12.47	231.71	215.67
LabelHallu	12.47	292.15	728.11
KT	4.91	32.69	865.56
DAATF	15.67	14.23	1001.66
SDA_CFM (ours)	**4.91**	**219.69**	**47.09**

6. Computational efficiency and deployment feasibility

Model complexity and inference latency are compared in Table X. SDA_CFM contains 4.91 M parameters, substantially fewer than RFS (12.47 M), LabelHallu (12.47 M), and DAATF (15.67 M). Its computational cost of 219.69 GFLOPs is higher than lightweight baselines such as MatchingNet and ProtoNet, but remains below LabelHallu and slightly below RFS. In terms of inference latency, SDA_CFM achieves 47.09 ms, far lower than KT (865.56 ms), LabelHallu (728.11 ms), and DAATF (1001.66 ms).

Overall, SDA_CFM offers a competitive trade-off among model size, computational cost, and inference speed, making it well-suited for practical deployment in resource-constrained underwater acoustic environments.

V. CONCLUSION

This paper proposes an SDA_CFM-based few-shot UATR framework to address the challenges of data scarcity and noise interference in complex marine environments. The proposed method combines SDA, TFA, CFM, and task-adaptive fine-tuning with PSG. The SDA module operates in the spatial-frequency domain of spectrograms to construct complementary multi-view samples by preserving coarse spectrotemporal structures while suppressing noise-sensitive fine texture variations. TFA is used to capture complementary dependencies along the temporal and frequency dimensions, while CFM performs lightweight guidance-based gated recalibration on augmented features. In addition, PSG enriches the support-set distribution during testing to improve adaptation to novel classes and cross-domain conditions.

The effectiveness of the proposed method is validated on the ShipsEar dataset under different shot and SNR settings. Experimental results show that SDA_CFM consistently outperforms the compared few-shot methods in both clean and low-SNR scenarios. In particular, under the 5-shot setting at SNR = –15 dB, the recognition accuracy reaches 51.00%. Systematic ablation studies further verify the contributions of the main components, showing that the proposed multi-view design, guided recalibration, and task-adaptive fine-tuning all contribute to improved robustness under noisy conditions.

Cross-domain experiments on the DanShip dataset further demonstrate the generalization ability of the proposed framework. In the clean 5-shot setting, SDA_CFM achieves an accuracy of 80.29%, and under the additional cross-domain noisy condition of SNR = –5 dB, it still maintains 67.59% accuracy. These results indicate that the proposed method remains effective under the combined challenges of domain shift, low SNR, and limited training samples.

However, the current study is conducted under a closed-set few-shot recognition setting, where all test samples are assumed to belong to known classes. This limits the applicability of the model in real-world scenarios involving background-only inputs or previously unseen categories. Extending the proposed framework to open-set or open-world recognition settings will be an important direction for future work.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts of interest to disclose.

DATA AVAILABILITY

The data that support the findings of this study are openly available in Santos-Domínguez *et al.* (2016) and Cui *et al.* (2025)

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